Recall January 2020

119 Question

PHYSIOLOGY:

1.The characteristic of a water-insoluble substance most important in governing its diffusibility through a cell membrane is its

a. Hydrated diameter

b. Molecular weight

c. Electrical charge

d. Lipid solubility

e. Three-dimensional shape

2.Which one of the following muscle proteins plays an important role in contraction of both smooth and striated muscle?

a. Calmodulin

b. Troponin

c. Tropomyosin

d. Actin

e. Myosin light chains

3.During the process of excitation-contraction coupling in skeletal muscle, calcium is released from the sarcoplasmic reticulum by

a. Inositol triphosphate (IP3)

b. Protein kinase A

c. An increase in intracellular calcium concentration

d. Membrane depolarization

e. An increase in intracellular sodium concentration

4.Which of the following words or phrases is most closely associated with an end-plate potential at the neuromuscular junction?

a. "All-or-none response"

b. Depolarization

c. Hyperpolarization

d. Action potential

e. Electrically excitable gates

5. In a nerve, the magnitude of the action potential overshoot is normally a function of the:

a. Magnitude of the stimulus

b. Intracellular potassium concentration

c. Extracellular sodium concentration

d. Resting membrane potential

e. Diameter of the axon

6. When comparing the contractile responses in smooth and skeletal muscle, which of the following is most different?

a. The source of activator calcium

b. The role of calcium in initiating contraction

c. The mechanism of force generation

d. The source of energy used during contraction

e. The nature of the contractile proteins

7. Nicotinic receptors are responsible for

a. Producing the skeletal muscle end-plate potential

b. Decreasing the rate of phase-4 depolarization at the SA node

c. Increasing the force of stomach contractions

d. Delaying the emptying of liquids from the stomach

e. Decreasing the excitability of sympathetic postganglionic neurons

8. The velocity of nerve conduction is increased with a decrease in the

a. Diameter of the nerve fiber

b. Degree of myelinization

c. Space constant of the nerve fiber

d. Capacitance of the nerve fiber membrane

e. Resting membrane potential

9. Which one of the following will be observed in a patient with chronic renal failure?

a. A decrease in the fractional excretion of sodium

b. An increase in the free water clearance

c. An increase in the anion gap

d. A decrease in the excretion of creatinine

e. An increase in the net acid excretion

10. Connexin is an important component of the

a. Gap junction

b. Sarcoplasmic reticulum

c. Microtubule

d. Synaptic vesicle

e. Sodium channel

11. In a test subject, oxygen consumption was measured at 700 mL/min. Pulmonary artery oxygen content was 140 mL per liter of blood and brachial artery oxygen content was 210 mL per liter of blood. Cardiac out- put was which of the following?

a. 4.2 L/min

b. 7.0 L/min

c. 10.0 L/min

d. 12.6 L/min

e. 30.0 L/min

12. Which one of the following is the best index of afterload?

a. Left ventricular end-diastolic pressure

b. Left ventricular mean systolic pressure

c. Pulmonary capillary wedge pressure

d. Total peripheral resistance

e. Mean arterial blood pressure

13. Which of the following would result from a regurgitant aortic valve in a nonfailing heart?

a. A decrease in diastolic pressure

b. A decrease in cardiac energy consumption

c. A systolic murmur

d.A decrease in heart rate

e.A decrease in systolic blood pressure

14. Stroke volume is increased by

a. A decrease in venous compliance

b. An increase in afterload

c. A decrease in contractility

d.An increase in heart rate

e.A decrease in coronary blood flow

15. Propagation of the potential through the heart is fastest in the

a. SA node

b. Atrial muscle

c. AV node

d. Purkinje fibers

e. Ventricular muscle

16. A patient in heart failure improves markedly after using a drug that increases the inotropic state of her heart. Which one of the following changes is primarily responsible for the improvement in her condition?

a.A reduction in heart rate

b.A reduction in heart size

c. An increase in end-diastolic pressure

d.An increase in wall thickness

e. An increase in cardiac excitability

17. Which one of the following is most likely to be observed in a patient with untreated atrial fibrillation?

a.An increased venous A wave

b. An increased left atrial pressure

c. A decreased heart rate

d. An increased stroke volume

e. An increased arterial blood pressure

18. Cardiovascular changes that occur during inspiration include decreased

a. Right ventricular filling

b. Right ventricular output

c. Pressure gradient from extrathoracic veins to the right atrium

d. Systemic blood pressure

e. Left ventricular contractility

19. During exercise, cardiac output is augmented by

a. Sympathetic stimulation of resistance vessels

b. Dilation of venous vessels

c. Decreased end-diastolic volume

d. Decreased mean systemic arterial pressure

e. Increased ventricular contractility

20. Which one of the following correctly describes an event that normally occurs during the PR interval?

a. The ventricle is contracting

b. The cardiac action potential passes through the AV node

c.There is no change in the voltage tracing on the ECG

d.The mitral and aortic valves are both closed

e.The second heart sound is heard

21. The consumption of oxygen by the kidney

a. Decreases as blood flow increases

b. Is regulated by erythropoietin

c. Remains constant as blood flow increases

d. Directly reflects the level of sodium transport

e. Is greatest in the medulla

22. In the presence of ADH, the filtrate will be isotonic to plasma in the:

a. Descending limb of the loop of Henle

b. Ascending limb of the loop of Henle

c. Cortical collecting tubule

d. Medullary collecting tubule

e. Renal pelvis

23. Sodium reabsorption from the distal tubule will be increased if there is an increase in

a. Plasma potassium concentration

b. Plasma volume

c. Mean arterial pressure

d. Urine flow rate

e. Plasma osmolality

24. Renin secretion by the kidney is increased by

a. Increasing mean blood pressure

b. Increasing glomerular filtration rate

c. Increasing sympathetic nerve activity

d. Increasing angiotensin II synthesis

e. Increasing atrial natriuretic hormone secretion

25. Which of the following is most likely to cause an increase in the glomerular filtration rate?

a. Contraction of mesangial cells

b. Blockage of the ureter

c. Release of renin from the juxtaglomerular apparatus

d. Dilation of the afferent arterioles

e. Volume depletion

26. The daily production of hydrogen ion from CO2 is primarily buffered by:

a. Extracellular bicarbonate

b. Red blood cell bicarbonate

c. Red blood cell hemoglobin

d. Plasma proteins

e. Plasma phosphate

27. Glomerular filtration rate would be decreased by

a. Constriction of the efferent arteriole

b. An increase in afferent arteriolar pressure

c. Compression of the renal capsule

d.A decrease in the concentration of plasma protein

e.An increase in renal blood flow

28. Renal correction of hyperkalemia will result in

a. Alkalosis

b. Acidosis

c. Increased secretion of HCO_3^-

d. Increased secretion of H^+

e. Increased excretion of Na^+

29. In metabolic acidosis caused by diabetic ketoacidosis, which of the following would be greater than normal?

a. Concentration of plasma HCO_3^-

b. Anion gap

c. Arterial PCO_2

d. Plasma pH

e. Blood volume

30. Which one of the following substances causes renal blood flow to decrease?

a. Nitric oxide

b. Bradykinin

c. Prostaglandins

d. Adenosine

e. Dopamine

MICROBIOLOGY:

1. Which of the following viruses causes an acute febrile rash and pro- duces disease in immunocompetent children but has been associated with transient aplastic crises in persons with sickle cell disease?

a. Rubeola

b. Varicella-zoster

c. Parvovirus

d. Rubella

e. Herpes simplex

2. Infection with herpes simplex virus, a common human pathogen, is best

described by which of the following statements?
a. The CNS and visceral organs are usually involved
b. It rarely recurs in a host who has a high antibody titer
c. It can be reactivated by emotional disturbances or prolonged exposure to sun- light
d. Initial infection usually occurs by intestinal absorption of the virus e. Infection with type 1 virus is most common

3. The latest and most effective therapy for AIDS patients includes azi-dothymidine (AZT), dideoxyinosine (DDI), and saquinavir or similar agents. Use of these three drugs would inhibit which of the following viral processes?
a. RNase, DNase
b. gp120 formation
c. p24 antibody expression
d. All membrane synthesis
e. Reverse transcriptase, protease

4. Infectious mononucleosis, a viral disorder that can be debilitating, is characterized by which of the following statements?
a. It is most prevalent in children less than 14 years old
b. It is caused by a rhabdovirus
c. The causative pathogen is an Epstein-Barr virus
d. Affected persons respond to treatment with the production of heter ophil anti-bodies
e. Ribavirin is the treatment of choice

5. Meningitis is characterized by the acute onset of fever and stiff neck. Aseptic meningitis may be caused by a variety of microbial agents. During the initial 24 h

of the course of aseptic meningitis, an affected person ś cerebrospinal fluid is characterized by

a. Decreased protein content

b. Elevated glucose concentration

c. Lymphocytosis

d. Polymorphonuclear leukocytosis

e. Eosinophilia

6. The presence of Negri inclusion bodies in host cells is characteristic of:

a. Mumps

b. Infectious mononucleosis

c. Congenital rubella

d. Aseptic meningitis

e. Rabies

7. A hospital worker is found to have hepatitis B surface antigen. Subse- quent tests reveal the presence of e antigen as well. The worker most likely

a. Is infective and has active hepatitis

b. Is infective but does not have active hepatitis

c. Is not infective

d. Is evincing a biologic false-positive test for hepatitis

e. Has both hepatitis B and C

8. Cytomegalovirus (CMV) infection is common. Which one of the following statements best characterizes CMV?

a. It can be transmitted across the placental barrier

b. While a common infection, CMV is almost always symptomatic

c. The CMV can be cultured from red blood cells of infected patients

d. Unlike other viral infections, CMV is not activated by immunosuppressive therapy

e. There is no specific therapy for CMV

9. Human rotaviruses are characterized by which of the following statements?

a. They produce an infection that is primarily seen in adults

b. They produce cytopathic effects in many conventional tissue culture systems
c. They are lipid-containing RNA viruses possessing a double-shelled capsid
d. They can be sensitively and rapidly detected in stools by the enzyme-linked immunosorbent assay (ELISA) technique
e. They have been implicated as a major etiologic agent of infantile respiratory dis- ease

10. Assuming that the average achievable serum level of gentamicin is 6 to 8 mcg/mL, which of the following bacteria is susceptible to gentamicin?
a. E. coli with a minimal inhibitory concentration (MIC) of 10 mcg/mL
b. E. coli with an MIC of 12 mcg/mL
c. Klebsiella with an MIC of 0.25 mcg/mL
d. Klebsiella with an MIC of 6.0 mcg/mL
e. Klebsiella with an MIC of 20 mcg/mL

11. A child comes to an emergency room because of an infected dog bite. The wound is found to contain small Gram-negative rods. The most likely cause of infection is
a. E. coli
b. H. influenzae
c. Pasteurella multocida
d. Brucella canis
12. Symptoms of C. botulinum food poisoning include double vision, in- ability to speak, and respiratory paralysis. These symptoms are consistent with
a. Invasion of the gut epithelium by C. botulinum
b. Secretion of an enterotoxin
c. Endotoxin shock
d. Ingestion of a neurotoxin
e. Activation of cyclic AMP

13. In people who have sickle cell anemia, osteomyelitis usually is associated with which of the following organisms?
a. Micrococcus

b. Escherichia

c. Pseudomonas

d. Salmonella

e. Streptococcus

14. The treatment of choice for a patient with C. jejuni enterocolitis is

a. Erythromycin

b. Ciprofloxacin

c. Ampicillin

d. Pepto-Bismol

e. Campylobacter antitoxin

15. A hyperemic edema of the larynx and epiglottis that rapidly leads to respiratory obstruction in young children is most likely to be caused by

a. K. pneumoniae

b. M. pneumoniae

c. Neisseria meningitidis

d. H. influenzae

e. H. hemolyticus

16. E. coli causes disease by a variety of different methods. Which one of the following E. colitypes is characterized by the presence of LT (heat-labile) and ST (heat-stable) proteins?

a. Enteroinvasive (EIEC)

b. Enterotoxigenic (ETEC)

c. Enterohemorrhagic (EHEC)

d. Enteropathogenic (EPEC)

e. Enterohemolytic (EHEEC)

17. Diphtheria toxin is produced only by those strains of C. diphtheriae that are

a. Glucose fermenters

b. Sucrose fermenters

c. Lysogenic for βprophage

d. Of the mitis strain

e. Encapsulated

18. The antibiotic of choice for Lyme disease is

a. Penicillin

b. Ampicillin

c. Erythromycin

d. Vancomycin

e. Ceftriaxone

19. A 2-year-old child was admitted to the hospital with acute meningitis. The Gram stain revealed Gram-positive short rods, and the mother indicated that the child had received "all" of the meningitis vaccinations. What is the most likely cause of the disease?

a. N. meningitidis, group A

b. N. meningitidis, group C

c. Listeria

d. S. pneumoniae

e. H. influenzae

20. The antibiotic therapy of choice for legionellosis is

a. Penicillin

b. Ampicillin

c. Erythromycin

d. Vancomycin

e. Ceftriaxone

21. N. gonorrhoeae is a fastidious pathogen and found in sites often contaminated with normal flora. The best medium for isolation is

a. Sheep blood agar

b. Löffler's medium

c. Thayer-Martin agar

d. Thiosulfate citrate bile salts sucrose medium

e. Löwenstein-Jensen medium

22. V. cholerae, the causative agent of cholera, is best isolated using

a. Sheep blood agar

b. Löffler ś medium

c. Thayer-Martin agar

d. Thiosulfate citrate bile salts sucrose medium

e. Löwenstein-Jensen medium

23. Patients should be vaccinated annually for influenza with which of the following vaccines?

a. Immune serum globulin

b. Killed virus vaccine

c. Inactivated virus vaccine

d. Live virus vaccine

e. Recombinant viral vaccine

24. The class of antibiotics known as the quinolones are bactericidal. Their mode of action on growing bacteria is thought to be?

a. Inhibition of DNA gyrase

b. Inactivation of penicillin-binding protein II

c. Inhibition of B-lactamase

d. Prevention of the cross-linking of glycine

e. Inhibition of reverse transcriptase

25. S. aureus causes a wide variety of infections, ranging from wound infection to pneumonia. Treatment of S. aureus infection with penicillin is often complicated by the

a. Inability of penicillin to penetrate the membrane of S.aureus

b. Production of penicillinase by S. aureus

c. Production of penicillin acetylase by S. aureus

d. Lack of penicillin binding sites on S. aureus protein

26. Pathogenic mechanisms involved in tuberculosis can be primarily attributed to which of the following?

a. Toxin production by the mycobacteria

b. Specific cell adhesion sites

c. Cell-mediated hypersensitivity

d. Humoral immunity

e. Clogging of alveoli by large numbers of acid-fast mycobacteria

27. Atypical lymphocytosis is most likely to be found in which one of the following diseases?

a. Encephalitis caused by herpes simplex virus (HSV)

b. Mononucleosis induced by Epstein-Barr virus

c. Parvovirus infection

d. Chronic hepatitis C

e. Rotavirus gastroenteritis

28. Transfer of antibiotic resistance between tow bacteria through abridge (sex pillus) is called:

a. transduction

b. transformation

c. conjugation

PATHOLOGY:

1. Osponization is the:

a) Formation of free radicals

b) Degradation of bacteria by lysozymes

c) Engulfment of antigen by leukocytes

d) Coating of antigen by antibodies

e) Processing of antigens by antigen presenting cells

2. Which of the following will predict a better prognosis for breast cancer:

a) The tumor is small in size

b) No metastasis are found in sampled lymph node

c) Numerous mitosis are seen

d) Tumor shows mild hyperchromatism

e) Tumor cells show marked pleomorphism

3. Multinucleated Giant Cells of the foreign body type origin from:

a) Nuclear Division of Granulocytes

b) Atypical Regeneration of Epithelium

c) Megakaryocytes

d) Fusion of Macrophages

e) Multiplication of Nuclei Surrounding Fibrocytes

4. Which of the following chemical mediators is most im
portant in the development of Granulomatous Inflammation:

a) Interferon gamma

b) Bradykinin

c) Complement C5a

d) Histamine

e) Prostaglandin E2

5. A purulent exudates is generally characterized by the presence of:

a) Mucous

b) Macrophages and connective tissue

c) Neutrophils and necrotic debris

d) Serous Fluid

e) Precipitated Protein

6. Epitheloid Cells and Langerhans Giant Cells in granulomas are derived from:

a) Neutrophils

b) Eosinophils

c) Mast Cells

d) Macrophage

e) Lymphocytes

7. Which of the following is true about an Exudate:

a) Occurs in heart failure

b) Is due to low plasma oncotic pressure

c) Has a low specific gravity

d) Has a high protein content

e) Is due to lymphatic obstruction

8. Fat emboli can be precipitated by:

a) fat necrosis

b) long bone fracture

c) deep vein thrombosis

9. Red Infarcts develop in:

a) Spleen

b) Liver

c) Kidney

d) Adrenal Glands

e) Intestine

10. Nephrotic syndrome:

a. High cholesterol level

b. Oliguria is a feature

c. mild proteinurea

d. high blood albumin

e. no swelling

11. injury inspite of restoration of blood supply?

Free radicals

12. Prostaglandins are formed from Arachidonic Acid through the action of which enzyme pathway?

a) Cyclooxygenase

b) Lipoxygenase

c) Myeloperoxidase

d) Phospholipase A

e) Glutathione Reductase

13. Which of the following cellular process is most likely to occur in the myocardium of a patient with myocardial infarction who is being treated with fibrinolytic agents:

a) Apoptosis

b) Free Radical Injury

c) Fatty Changes

d) Accumulation of Lipofuscin

e) Accumulation of Amyloid

14. An Example of metastatic Calcification is seen in:

a. Sarcodiosis

b. TB

c. Benign prostatic hyperplasia

15. Which of the following is TRUE about fatty change:

A. Increased triglyceride in parenchyma

B. Most common cause is DM

C. Occurs in the breast

D. Occurs during pregnancy due to accumulation of fetal fat in maternal blood

16. Definite sign of necrosis:

a) Glycogen stores are depleted

b) Cytoplasmic sodium increases

c) Nuclei undergo karyorrhexis

d) Intracellular pH diminishes

e) Blebs form on cell membranes

17. Enzyme that increase due to liver damage is :

a. GoD

b. GGT

c. ALP

d. Bilirubin

e. Serine hydroxyamelase

18. Pancreatic necrosis:

A. Coagulative

B. Fat necrosis

C. Collaguitive

19. A question about lukocyte adhesion:

a. Interferon gamma

b. Selectin

c. Prostacyclin

20. Staging of cancer is according to:

a. Differentiation of cells

b. Metastasis

c. Invasion of tumor

d. Affected lymph node

e. Tumor size

ANATOMY:

1. In a 50 years old female blood supply to the head of femur is from:

a. Profunda femoris artery

b. Branches from Obturator artery

c. From the marrow

d. Retinacular arteries

e. Superior gluteal

2. Pt diagnosed as ovarian carcioma came complain of medial side of the thigh parathesia which nerve is affected:

a. Femaral trunk

b. Obtorator nerve

c. Medial cutaneous nerve if the thigh

d. Lumbosacral trunk

e. Intermediate cutaneous nerve of the thigh

3. Muscle not attached to perineal body

a. Levator ani

b. Puporectalis

c. Bubovaginalis

d. Transverse perineal muscle

e. Ex anal sphincter

4. Femoral pulsation is detected at which point

a. Midpoint in inguinal ligament

b. 2/3 away from the pubic tubercle)

c. Mid inguinal point

d. Between the aSIS and pupic tubercle

e. At the deep inguinal ring

5. Venous drainge of the lower limbs is mostly go to

a. Great suphenous vein

b. Deep veins

c. Femoral vein

d. Internal iliac vein

6. Cancer from pelvic viscera can deposits cancerous cells in the vertebrae this communication occur between the internal iliac veins and the vertebral veins through the:

a. Lateral sacral veins

b. Lumbar vein

c. Prostatic plexus

d. Umbilical veins

e. Pudenal vein

7. A 2 years baby had hernial sac medial and above the pubic tubercle, this is

a. Direct inguinal hernia

b. Indirect inguinal hernia

c. Femoral hernia

d. Hydrocele

8. Patient had abcess in the gluteal region. What is the Lymphayic drainge of the gluteal region:

a. Deep group

b. Internal iliac LNs

c. Lateral group of superficial inguinal

d. Medial proximal inguinal

9. In erect position, fluid in abdominal cavity will collects in

a. Subhepatic

b. Pouch of douglas

c. Rt Hepatorenal

10. deep perineal pouch in male contain

a. Prostatic urethra

b. bulbourethral gland

c. Isciocavernosus muscle

11. Ischiorectal fossae both communicate:

a. Posteriorly

b. anteriorly

c. Superiorly

d. Medialy

e. Lateral

12. Inferior wall of hunter canal

a. Adductor longus

b. Adductor magnus

c. Adductor previs

d. Gracillis

e. Obtorator externus

13. Muscle of urogenital diaphragm include

a. Sphincter urethrae

b. Sphincter vesicae

c. Levator ani

d. Ischiocavernosus

e. Obturator internus

14. Concerning the vas deferens, all of the following statements are true except:

a) It emerges from the deep inguinal ring and passes around the lateral margin of the inferior epigastric artery

b) It crosses the ureter in the region of the ischial spine

c) The terminal part is dilated to form the ampulla

d) It lies on the posterior surface of the prostate but is separated from it by peritoneum

e) It joins the duct of the seminal vesicle to form the ejaculatory duct

15. Regarding the anterior abdominal wall:

a. Superficial tissues of the anterolateral wall above theumbilicus to axillary nodes

b. Intertubercular plane lies at the level of L4

c. Mid-inguinal point lying halfway between the pubic tubercle and Anterior Superior Iliac Spine

d. Posterior wall of the inguinal canal is formed by the transversalis muscle

e. Sugrical incision at linea alba heals faster than that along paramedian abd wall

16. Concerning urinary bladder

a. Against fascia of levator ani

b. Detruser muscle supplied by sympathetic motor fibre

c. Connected to remnant of cloaca

d. In female is completely pelvic organ

e. Its sphincter vesicae has parasympathetic motor supply

17. Concerning the peritoneum

a. Right sub-hepatic pouch also called the lesser sac

b. The medial umbilical ligament contains the urachus

c. It ś a mucous membrane

d. Left sub-hepatic pouch is the hepatorenal pouch of morris

e. The epiploic foramen is bounded anteriorly by the portal vein

18. Question about anal canal

19. Krukenberg Tumor of the Ovary is an example of:

a) Lymphatic Spread

b) Vascular Spread

c) Transcolomic Spread

d) Intraductal Spread

e) Transluminal Spread

20. Hip joint is characterized by a huge capsule. Which of the following is attached to it posteriorly:

a. Greater trochanter

b. Intertubercular crest

c. Acetabulum

d. Intertubercular line

BIOCHEMISTRY:

1. The following are purine nucleotide

a. CTP and GDP

b. NAD and TDP

c. ATP and CMP

D. GMP and ADP

2/Which of the following amino acids is glucogenic but not ketogenic;

a. Tyrosine

b. Tryptophan

c. Valine

d. Phenylalanine.

e. Isoleucine

3. Beriberi?

a. is not common in alcoholics

b. the deficient vitamin never affect the brain

c. the deficiency of riboflavin

d. affect the nerves and the heart

e. is the combination of diarrhea ,dementia, dermatitis

4. During prolonged fasting:

A. B-oxidation of fatty acids is inhibited

B. Glycogenesis is very active

C. Blood glucose level is maximum

D. Ketogenesis is maximal

E. Bicarbonate is high

5. Synthetic nucleotide analogs?

a. Can inhibit DNA synthesis

b. Are always modified purine nucleotides

c. Lack phosphate group

d. Allow replicaton to proceed at faster rate

e. They cause denaturation of DNA

6/All the following use ketone bodies oxidation as significant source of energy except:

a. Liver

b. Kidney

c. RBC

d. Heart

e. Skeletal muscle

7/ Arachidonic acid?

a. Is branched fatty acid

b. Contains 3 double bond

c. Is a precursor for eicosanoids

d. Contain 5 double bonds

e. Is a20-carbon saturated fatty acid

8. Hypoxic injury in cells occur due to:

a. Reduction in ATP

b. Low blood

c. Glycogen depletion

d. Na leak outside cell

e. Decrease anaerobic glycolysis

9. The bonds that stabilize the structure of DNA are:

a. Hydrogen bonds and peptide bonds

b. Phosphodiester bonds and hydrogen bonds

c. Glycosidic bonds and thioester bonds

d. Disulphide bonds and hydrogen bonds

10. A Lipoprotein lipase:

a. Hydrolyses cholesteryl ester

b. Is present in the cytosol of adipocytes

c. Is active during starvation

d. Hydrolyses TAG in chylomicrons and VLDL

e. Is activated by glucagon

11/Waston and crick model of DNA structure implied that:

a. Thymine base pairs with adenine

b. Thymine base paires with cytosine

c. DNA is a single strand

d. The purine content is not equal to the pyrimidine content

e. The nitrogenous bases from the backbone of the molecule

12. Lactic acidosis caused by deficiency of?

a. Thymine

b. B1

c. Folic acid

d. Biotin

13. Which of the following statement about galactosemia is correct?

a. treatment involves restriction of glucose in the diet

b. A deficiency of glucokinase is the major cause

c. Is treated by elimination of galactose from the diet

d. Can be treated by restricting sucrose in the diet

e. Is inherited as autosomal dominant character

14. Pyruvate carboxylase

a. Converts acetyl CoA to pyruvate

b. Requires carnitine

c. Converts pyruvate to oxaloacetate

d. Is activated by high fructose 1,6 bisphosphate

e. Requires pyridoxal phosphate

15. Nucleosome:

a. Contain negatively charged protein

b. Is coiling of each single strand of DNA and histone

c. Is association of each DNA and arginine rich protein

d. Is combination of DNA and chromatin

e. Is association of DNA ,RNA,and non histone protein

16. All of the following are consumed in urea cycle except :

a. aspartate

b. Co2

c. Ornithine

d. NH3

e. ATP

17. Which clinical laboratory observation below is suggestive of Hartnup disease

(neutral amino acid transport deficiency)?

a. Burnt-sugar smell in urine

b. High plasma phenylalanine levels

c. Extremely high levels of citrulline in urine

d. Elevation of glutamine in blood and urine

e. Elevated plasma tyrosine and methionine levels

f. Dark urine

g. High fecal levels of tryptophan and indole derivatives

18. Cholesterol :

a. Normal cell membrane constituent

b. Precursor of sphingilipids

c. Main source of energy in the body

d. Produced by HMG lyase

19. Synthesis of glucose from pyruvate by gluconeogenesis in the liver:

A. Occurs exclusively in the liver

B. Is inhibited during prolonged fasting

C. Requires participation of Biotin

D. Requires lactate as intermediate

E. Occurs exclusively in the mitochondria

20. Question about glutamate

STATISTICS:

1. A drug is administered to 3 groups of healthy volunteers to evaluate the therapeutic dose and followed by measuring the medication excreted in urine in scheduling time frame, this type of is?

a. ANOVA

b. ...

2. Following cigarette smokers for 20 years to evaluate development of ca lung,

heart did ...etc. What is the type this study

a. Cohort study

b. Cross sectional

c. Case control

إعدادوتنسيق: د أسرار تاج السر
ومجموعة(بإذن الله ناجحين)